Attendance Code: 891876



UNIVERSITY OF LINCOLN

CMP2028M

Natural Language Processing

Week 8: “Information Extraction”

Heriberto Cuayahuitl

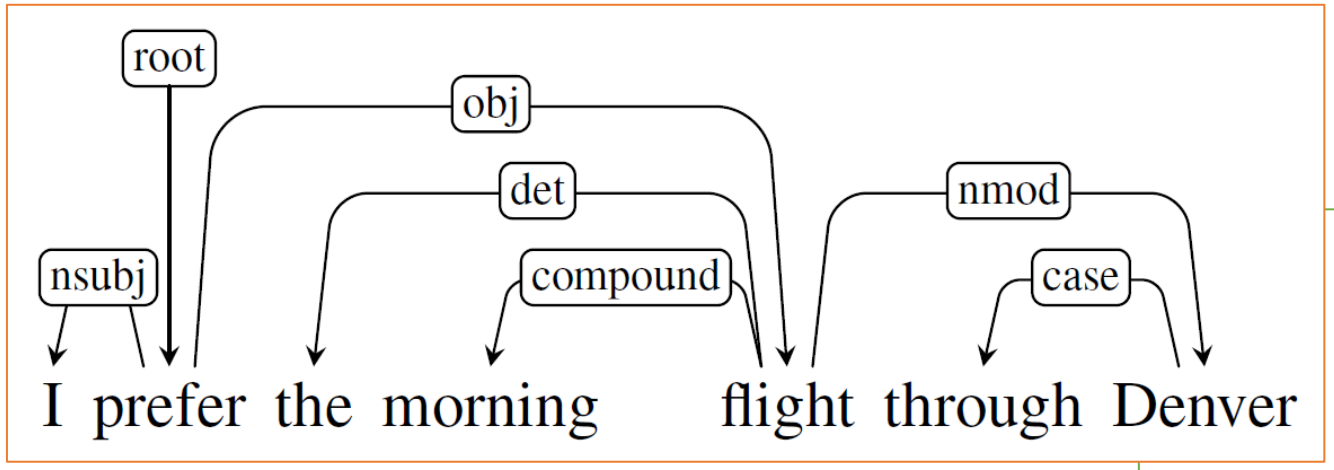
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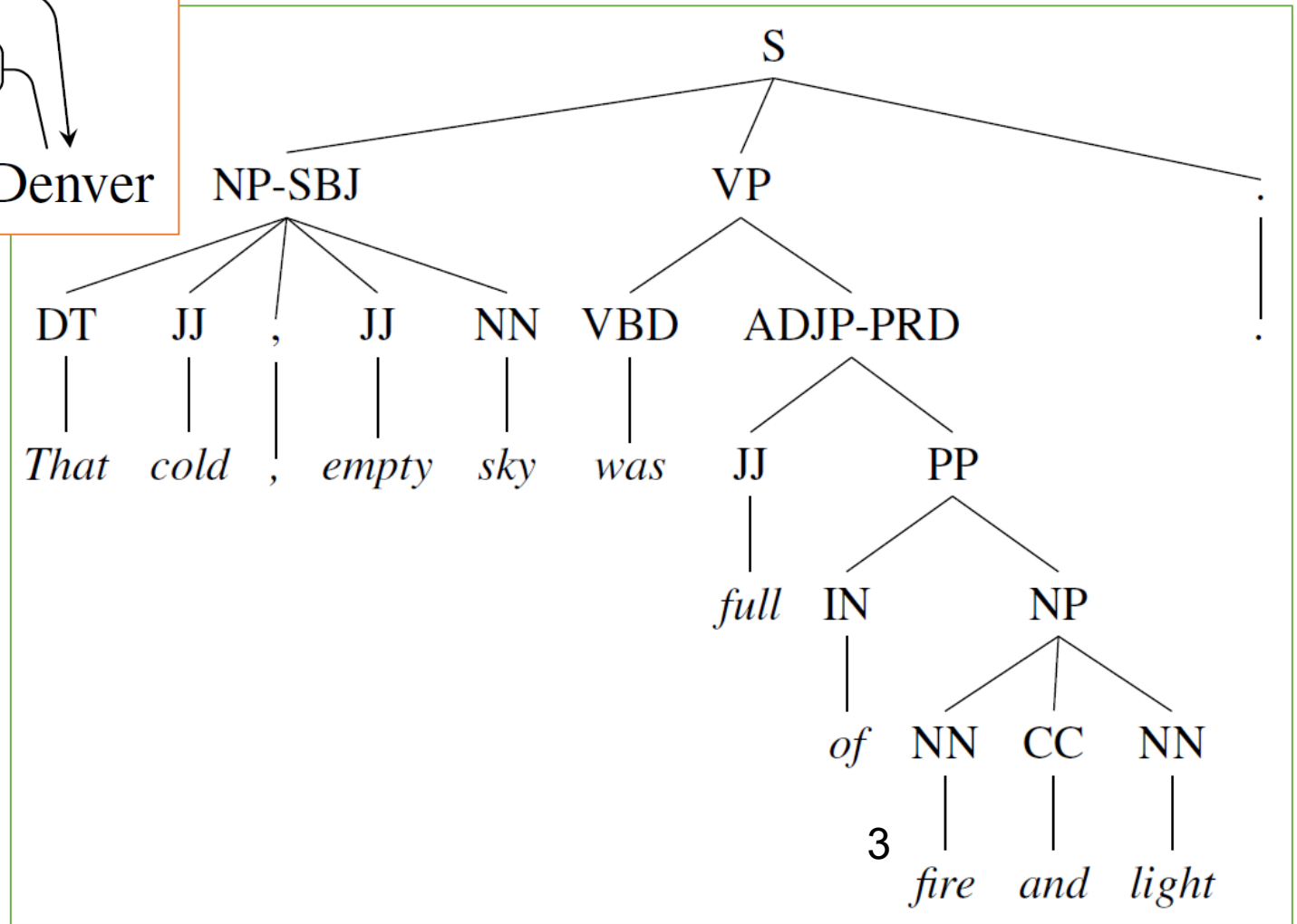
Agenda

- Week 1, 5-6 Feb: Tokenization and regular expressions
- Week 2, 12-13 Feb: N-gram language models
- Week 3, 19-20 Feb: Text classification
- Week 4, 27-27 Feb: Vector semantics and embeddings
- Week 5, 5-6 Mar: Large Language Models (LLMs)
- Week 6, 12-13 Mar: Question answering & information retrieval
- Week 7, 19-20 Mar: Constituency and dependency parsing
- Week 8, 26-27 Mar: **Information extraction**
- Week 9, 23-24 Apr: Overview of neural networks
- Week 10, 30 Apr-1May: Machine translation
- Week 11, 7-8 May: Chatbots and dialogue systems
- Week 12, 14-15 May: Speech recognition and synthesis

Easter break

Last Week: Dependency & Constituency Parsing





Today

- **Named Entity Recognition**

- Three Tasks:
  - Relation Extraction
  - Event Extraction
  - Time Extraction

Acknowledgement: Some images from https://web.stanford.edu/~jurafsky/slp3/9.pdf

Named Entity Recognition (NER)

Citing high fuel prices, [ORG **United Airlines**] said [TIME **Friday**] it has increased fares by [MONEY **$6**] per round trip on flights to some cities also served by lower-cost carriers. [ORG **American Airlines**], a unit of [ORG **AMR Corp.**], immediately matched the move, spokesman [PER **Tim Wagner**] said. [ORG **United**], a unit of [ORG **UAL Corp.**], said the increase took effect [TIME **Thursday**] and applies to most routes where it competes against discount carriers, such as [LOC **Chicago**] to [LOC **Dallas**] and [LOC **Denver**] to [LOC **San Francisco**].

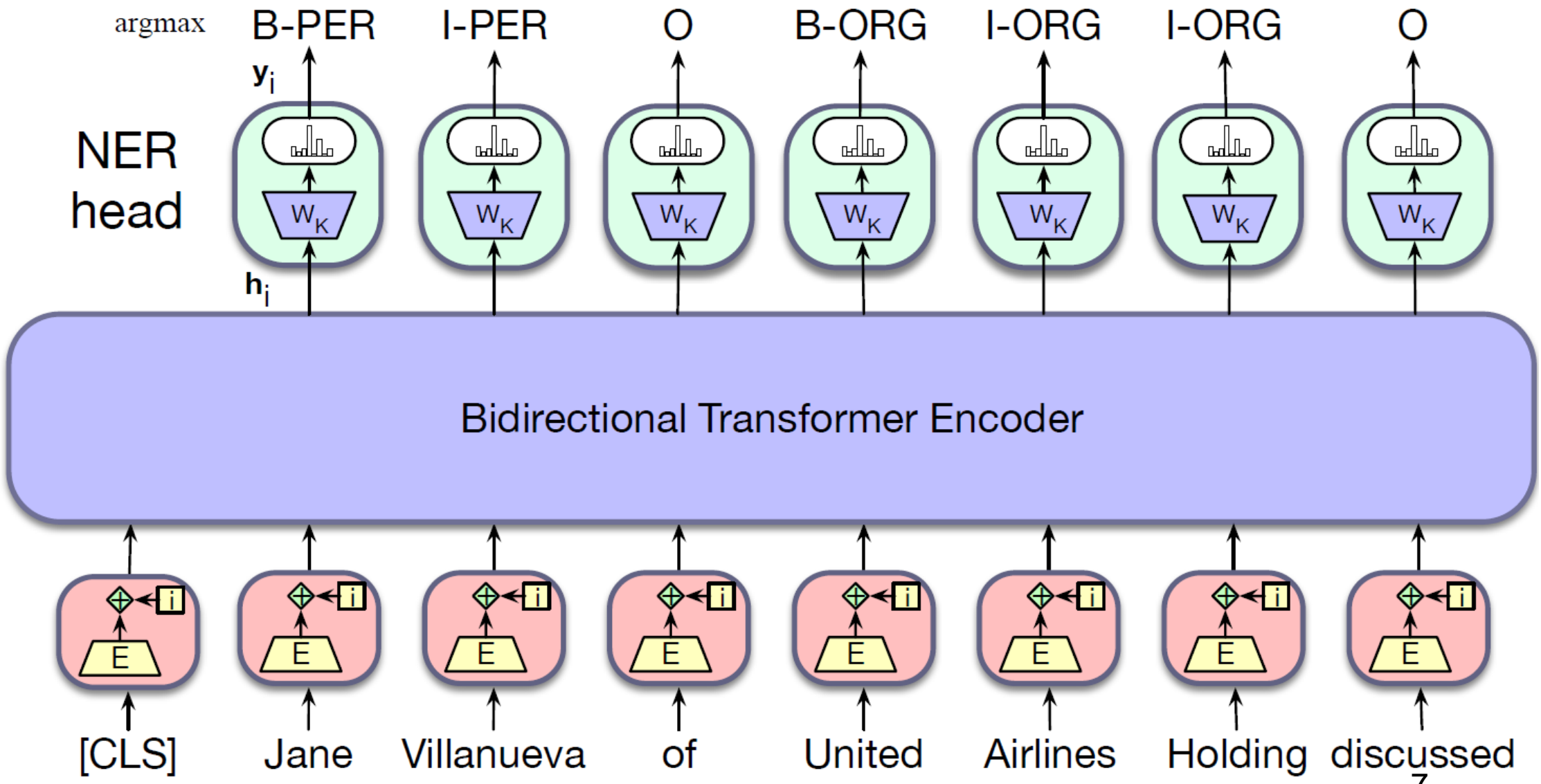
- A **named entity** is anything that can be referred to with a proper named entity name: a person, a location, an organisation.
- The task of a **NER system** is to find spans of text that constitute proper names and tag the type of the entity.

Bio Tagging and Variants

Words	IO Label	BIO Label	BIOES Label
Jane	I-PER	B-PER	B-PER
Villanueva	I-PER	I-PER	E-PER
of	O	O	O
United	I-ORG	B-ORG	B-ORG
Airlines	I-ORG	I-ORG	I-ORG
Holding	I-ORG	I-ORG	E-ORG
discussed	O	O	O
the	O	O	O
Chicago	I-LOC	B-LOC	S-LOC
route	O	O	O
.	O	O	O

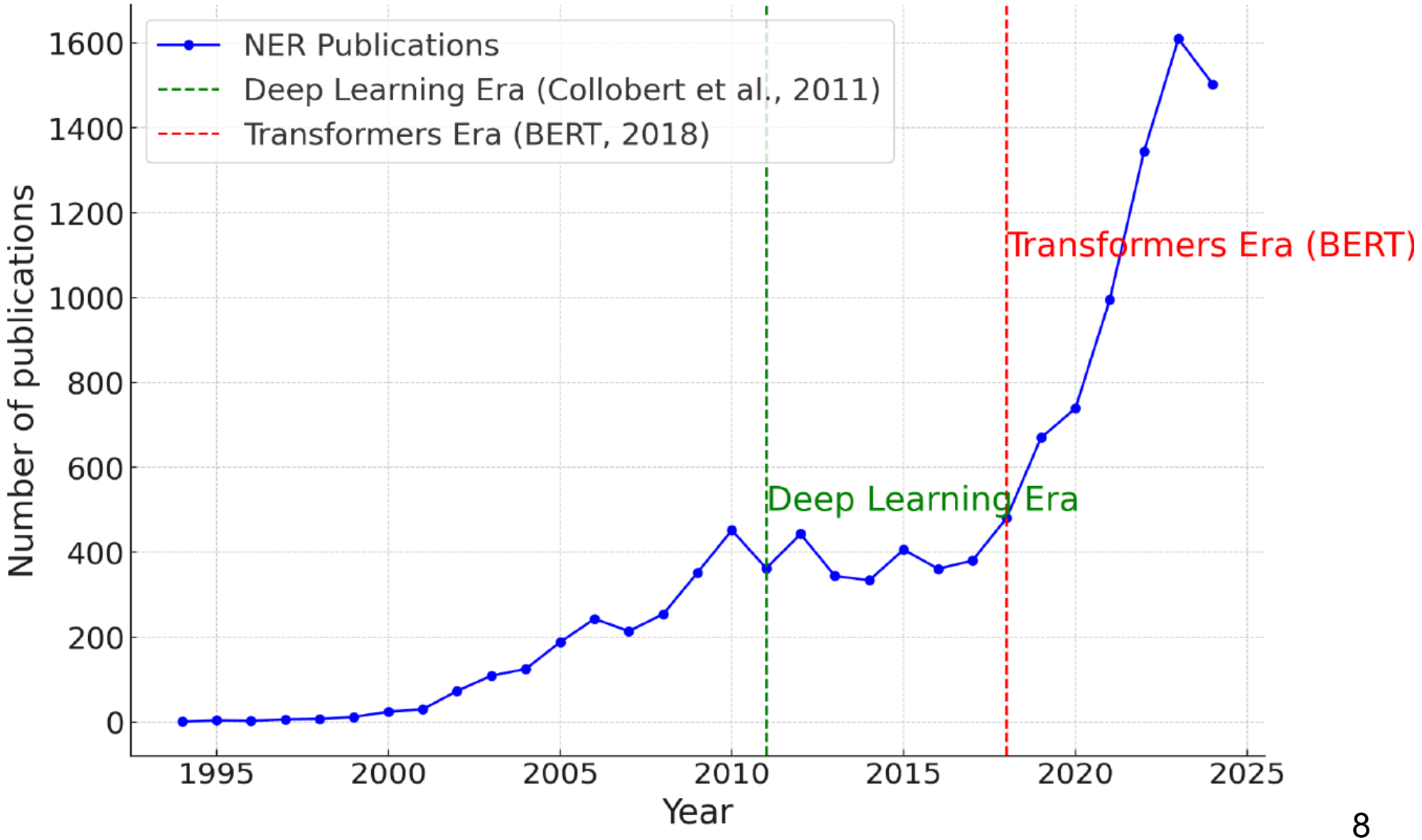
Bio Tagging allows us to treat NER like a word-by-word sequence labeling task, via tags (B=begin, I=inside, O=outside) that capture both the boundary and the named entity type.

Transformers for NER



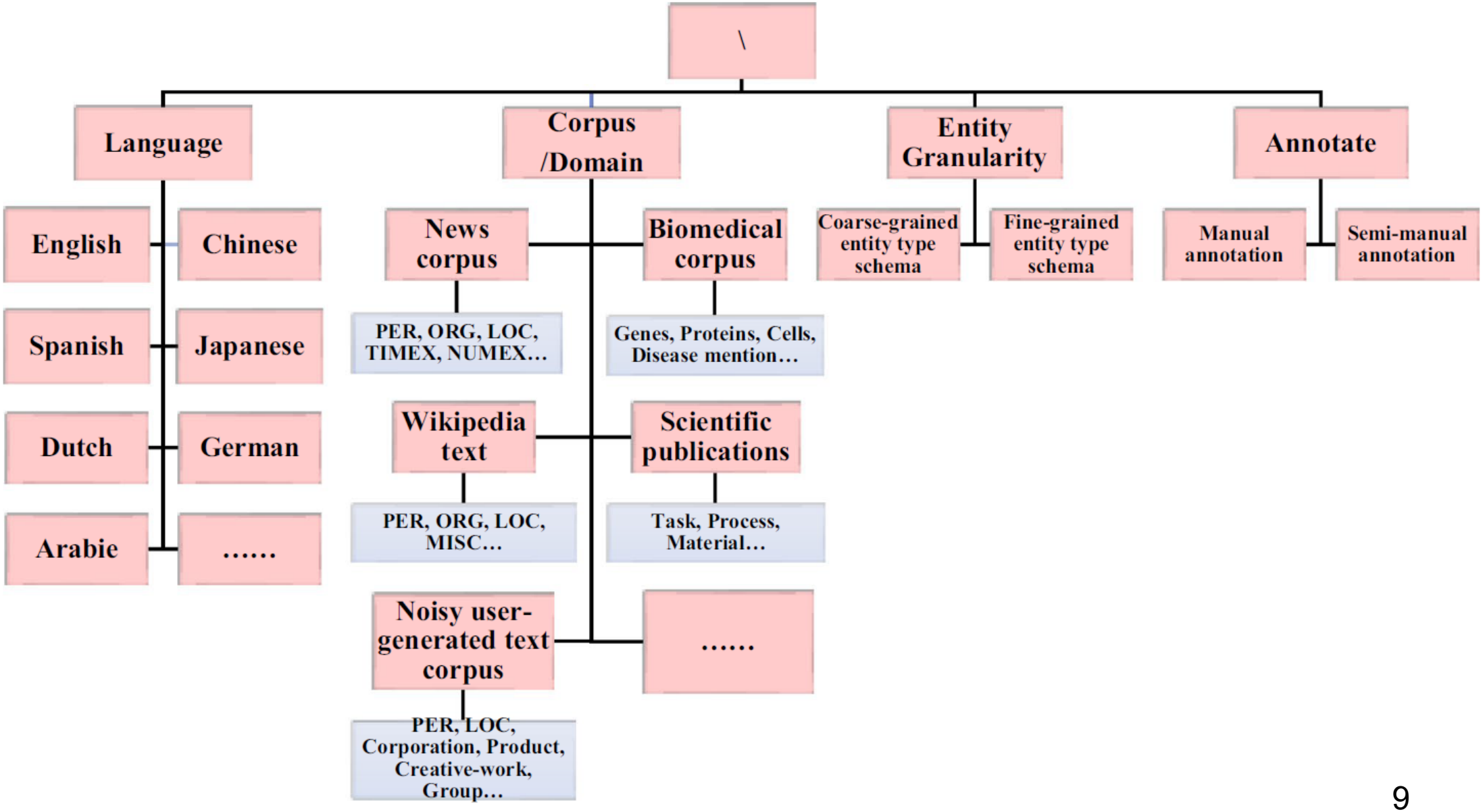
NER, similarly to Part-of-Speech (tagging) use evaluation metrics for text classification – see slides of week 9.

Transformers for NER



Source: Keraghel, et al. ``Recent Advanced in Named Entity Recognition: A Comprehensive Survey …, CoRR, 2024.

NER Datasets



Source: Keraghel, et al. ``Recent Advanced in Named Entity Recognition: A Comprehensive Survey …, CoRR, 2024.

Today

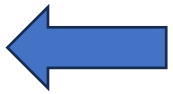
- Named Entity Recognition

- Three Tasks:
  - **Relation Extraction**
  - Event Extraction
  - Time Extraction

Acknowledgement: Some images from https://web.stanford.edu/~jurafsky/slp3/20.pdf

Semantic Relations from NERs

Citing high fuel prices, [ORG **United Airlines**] said [TIME **Friday**] it has increased fares by [MONEY **$6**] per round trip on flights to some cities also served by lower-cost carriers. [ORG **American Airlines**], a unit of [ORG **AMR Corp.**], immediately matched the move, spokesman [PER **Tim Wagner**] said. [ORG **United**], a unit of [ORG **UAL Corp.**], said the increase took effect [TIME **Thursday**] and applies to most routes where it competes against discount carriers, such as [LOC **Chicago**] to [LOC **Dallas**] and [LOC **Denver**] to [LOC **San Francisco**].



NERs

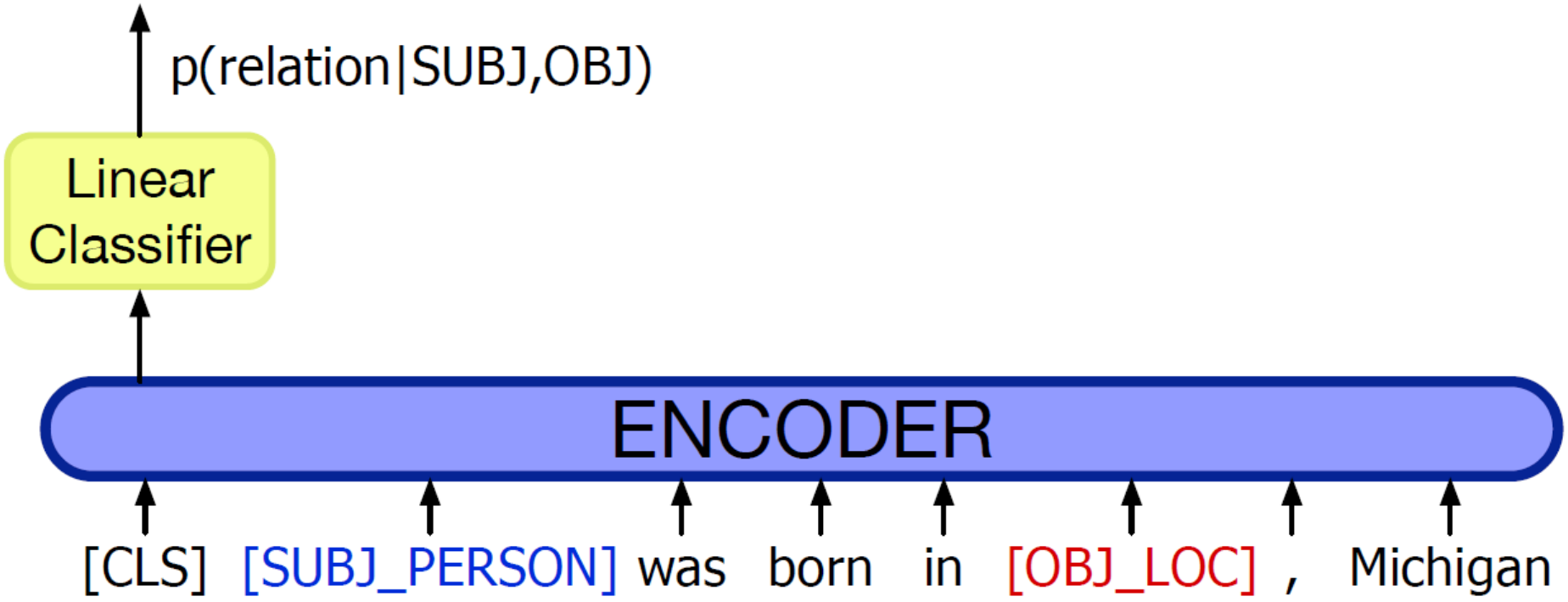
Relations	Types	Examples
Physical-Located	PER-GPE	**He** was in **Tennessee**
Part-Whole-Subsidiary	ORG-ORG	**XYZ**, the parent company of **ABC**
Person-Social-Family	PER-PER	**Yoko**'s husband **John**
Org-AFF-Founder	PER-ORG	**Steve Jobs**, co-founder of **Apple**...

Semantic Relations: Anoter Example

Example	Entities & Relations
Carey will succeed Cathleen P. Black, who held the position for 15 years and will take on a new role as chairwoman of Hearst Magazines, the company said.	PERSON/TITLE Relation=**per:title**
Irene Morgan Kirkaldy, who was born and reared in Baltimore, lived on Long Island and ran a child-care center in Queens with her second husband, Stanley Kirkaldy.	PERSON/CITY Relation=**per:city of birth**
Baldwin declined further comment, and said JetBlue chief executive Dave Barger was unavailable.	PERSON/TITLE Relation=**no relation**

Zhang, Y., et al. Position-aware attention and supervised data improve slot filling. EMNLP, 2017.

Relation Extraction via Text Classification



Labelled data (e.g., TACRED dataset) is required for this task!

Zhang, Y., et al. Position-aware attention and supervised data improve slot filling. EMNLP, 2017.

Relation Extraction via Text Classification: Algorithm

function FINDRELATIONS(*words*) **returns** *relations*

relations ← *nil*
entities ← FINDENTITIES(*words*)
forall entity pairs ⟨*e1*, *e2*⟩ **in** *entities* **do**
  if RELATED?(*e1*, *e2*)
    relations ← *relations*+CLASSIFYRELATION(*e1*, *e2*)

Recall that text classifiers can use features at different levels of granularity, from lexical to syntactic to embeddings.

Note, the classification approach to relation extraction is only one of the possible approaches. Other approaches include semi-supervised learning, distant supervision and unsupervised extraction.

Evaluation of Relation Extraction

General metric:

$$Precision = \frac{\text{\# of correctly extracted relation tuples in the sample}}{\text{total \# of extracted relation tuples in the sample.}}$$

Other metrics (from text classification, week 3+):

- Classification/Balanced Accuracy
- Precision-Recall, F1-Score
- Exact Match

See Section 8 of Keraghel, et al. ``Recent Advanced in Named Entity Recognition: A Comprehensive Survey …, CoRR, 2024.

Today

- Named Entity Recognition

- Three Tasks:
  - Relation Extraction
  - **Event Extraction**
  - Time Extraction

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Event Extraction

[EVENT Citing] high fuel prices, United Airlines [EVENT said] Friday it has [EVENT increased] fares by $6 per round trip on flights to some cities also served by lower-cost carriers. American Airlines, a unit of AMR Corp., immediately [EVENT matched] [EVENT the move], spokesman Tim Wagner [EVENT said]. United, a unit of UAL Corp., [EVENT said] [EVENT the increase] took effect Thursday and [EVENT applies] to most routes where it [EVENT competes] against discount carriers, such as Chicago to Dallas and Denver to San Francisco.

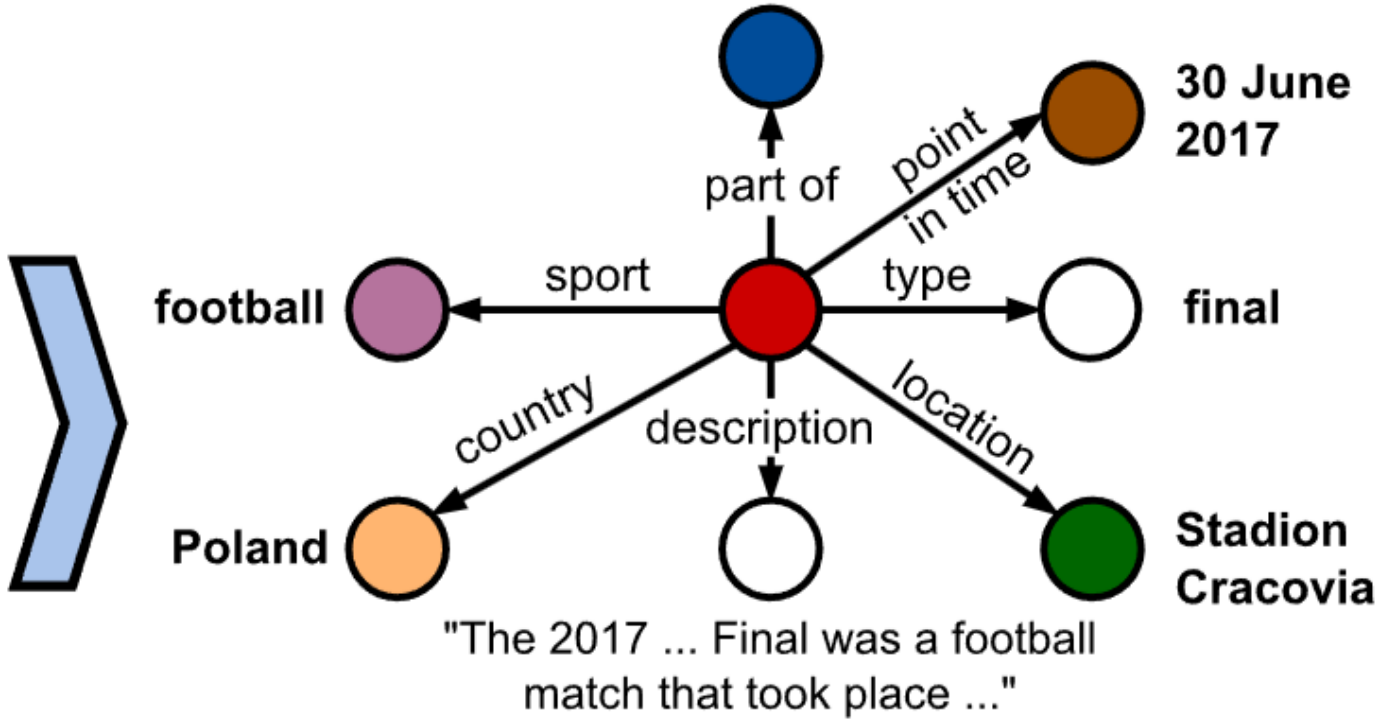
- Event extraction consists of identifying mentions of events in texts.
- An event mention is any expression denoting an event or state that can be assigned to a particular point, or interval, in time.

Event Extraction: Example

The 2017 UEFA European Under-21 Championship Final was a **football** match that took place on **30 June 2017** at the **Stadion Cracovia** in **Poland** to determine the winners of the **2017 UEFA European Under-21 Championship**.

Queries:

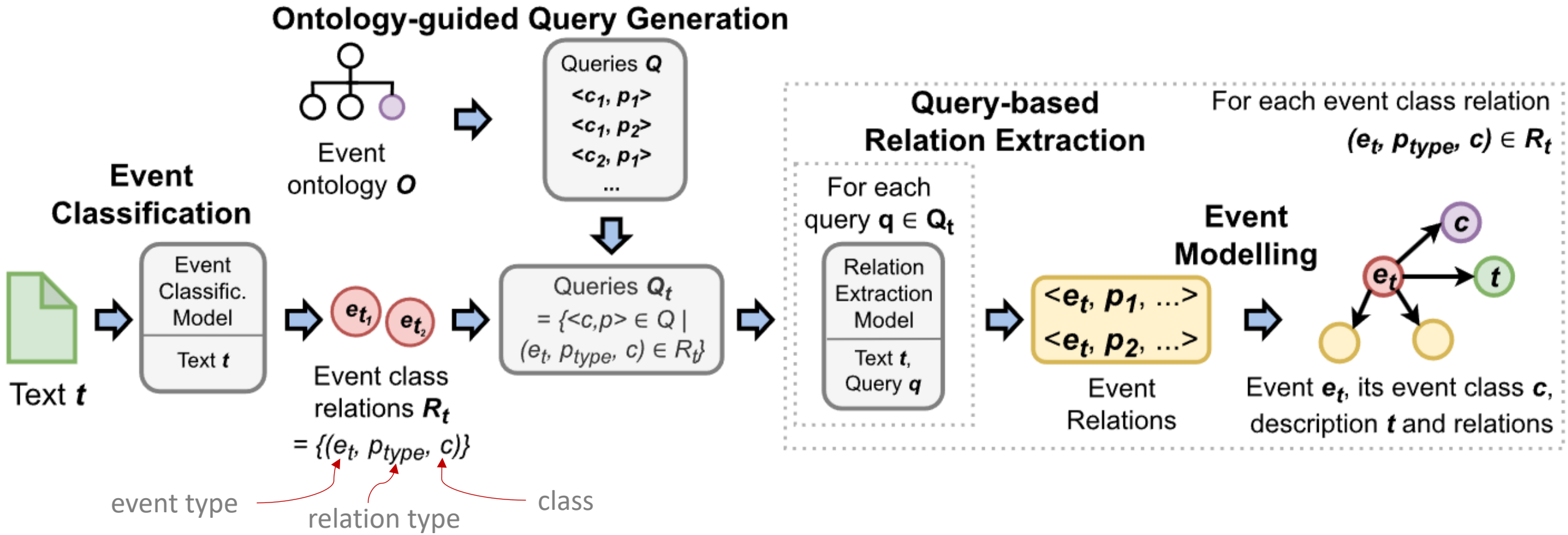
<final, sport>
<final, point in time>
<final, location>
<final, country>
<final, part of>



Event Extraction aims to identify & classify events and their relations in text, i.e., an event can be represented as a graph of entities and relations.

Kukulo, et al. "Transformer-Based Architectures Versus LLMs in SemanticEvent Extraction: Evaluating …". Semantic Web, 2025.

Event Extraction via Text Classification



The pipeline combines an event classification model with a query-based relation extraction model to construct a structured event graph from text.

Kukulo, et al. "Transformer-Based Architectures Versus LLMs in SemanticEvent Extraction: Evaluating …". Semantic Web, 2025.

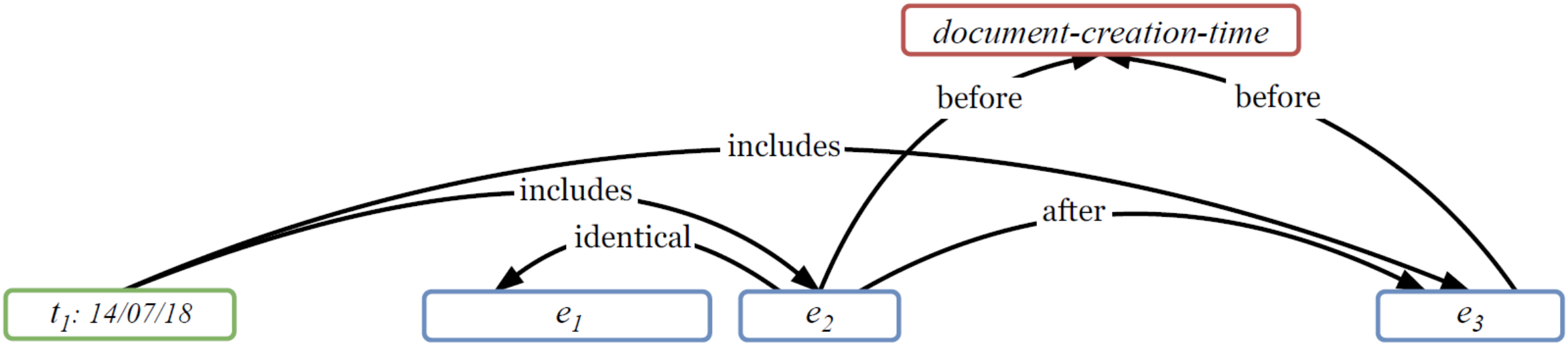
Today

- Named Entity Recognition

- Three Tasks:
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  - Event Extraction
  - **Time Extraction**

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Temporally Annotated Text



On July 14, a violent demonstration erupted in Chicago after a police shooting.

In time extraction, we typically want to work with time expressions and events, not just "time events".

Leeuwenberg & Moens. "A Survey on Temporal Reasoning for Temporal Information Extraction from Text". JAIR, 2019.

Today

- **Named Entity Recognition**: a text classification task that identifies and labels spans of text as predefined entity types such as people, organizations, and locations.

- **Relation, Event and Time Extraction**: They build on NER to capture meaning in text. Relation extraction identifies relationships between entities, event extraction detects events along with their participants and attributes, and time extraction identifies temporal expressions and links them to events to determine when they occur.

What's Next?

- Readings:
  - Jurafsky and Martin. Speech and Language Processing. Chapter 20, 2026. [recommended]
  - Keraghel, et al. ``Recent Advanced in Named Entity Recognition: A Comprehensive Survey …", CoRR, 2024.

- Workshop **tomorrow**: Information extraction code.

- Next lecture: Overview of Neural Networks